

Beyond the Dyadic Fabius Web

A geometric q -Fabius–Rvachev family, endpoint laws, zero arithmetic, Bernoulli–Bell polynomial calculus, and phase-aware extrapolation

Frontier research report based on the complete ProveIt
Fabius documentation corpus

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Abstract

The dyadic Fabius function, its inverse, Rvachev’s compactly supported up-function, and the signed Thue–Morse extension form an unusually dense network of exact identities. The inspected repository already develops that network at considerable depth: exact dyadic values, Gaussian-binomial coefficient extraction, Bernoulli and Bell polynomial formulas, Lambert- W endpoint phases, geometric Lagrange interpolation, Richardson filters, and an extensive campaign on the decay of the infinite sinc product. This report therefore does not attempt another rearrangement of those results. Instead it moves in a transverse direction by replacing the dyadic ratio $1/2$ by a free geometric parameter $0 < q < 1$.

For independent $U_j \sim \text{Unif}(0, 1)$, define

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad F_q(x) = \mathbb{P}(X_q \leq x), \quad Y_q = 2X_q - 1.$$

At $q = 1/2$ this is exactly the bounded Fabius distribution, and the density of $Y_{1/2}$ is Rvachev’s up-function. For the full family we prove a refinement equation and a rapidly decaying infinite sinc product; exact Bernoulli cumulants and Bell moments; a Bernoulli–Appell polynomial inverse to geometric uniform averaging; a finite-stage q -binomial scale identity; monotone contraction in convex order; an exact shape phase transition with a central plateau precisely when $q < 1/2$; a Gaussian degeneration as $q \uparrow 1$ with explicit standardized cumulants; a two-term logarithmic endpoint law

$$\log F_q(x) = -\frac{\log^2(1/x)}{2 \log(1/q)} - \frac{\log(1/x) \log \log(1/x)}{\log(1/q)} + \mathcal{O}_q(\log(1/x));$$

and its inverse Lambert- W quantile law. We also determine Fourier-zero multiplicities for reciprocal-integer ratios, isolate the arithmetic condition for cross-scale collisions, and explain why base two alone converts the finite product into a contiguous Thue–Morse block. Finally, we give a phase-preserving Lagrange/Richardson filter for asymptotic remainders whose coefficients are periodic in a lower-Lambert phase, together with a Fourier phase-averaging filter and exact Lambert- W node construction.

A post-snapshot Lean audit now supplies the generic continuous-linear naturality and conditional operator/scalar affine-recursion engine for arbitrary norm-summable weighted uniform series. It also defines this report’s geometric- q weights, series, and law and proves their measure-level self-similarity, reflection, and carried-unit-interval results. It does not prove equality of topological support, atomlessness, CDF or density identities, transforms, moments or cumulants, or fixed-point uniqueness.

The word *new* is used only in a corpus-relative sense: these statements were not found in the 57 TeX sources inspected in the specified repository snapshot. No claim of global priority is made. Proven statements are separated sharply from conjectural transseries, strict log-concavity, Edgeworth, inverse-phase, and arithmetic-collision problems. Deterministic numerical experiments and all plotting code accompany the report.

Status convention. A theorem or proposition in this report is accompanied by an argument intended to be mathematically complete at the paper level. A conjecture is explicitly labeled and is not asserted as proved. None of the new statements has been formalized in Lean as part of the repository snapshot. The principal novelty filter was comparison against all 57 TeX files listed in `corpus_inventory.txt`, including frozen versions, the living syntheses, the Thue–Morse atlas, all included paper sources, and the twelve Fourier-decay drafts and audits.

Since that pinned snapshot, `WeightedUniformSeries.lean` and `WeightedUniformDistribution.lean` have formalized the reusable product space, absolutely summable Banach-valued uniform-coordinate series, and their pushforward laws, including the eleven-declaration generic naturality and operator-affine recursion tranche, the compatibility API, reflection, and interval concentration. The specialization `GeometricUniformLaw.lean` defines $w_j(q) = (1 - q)q^j$ and X_q for every real q , proves absolute convergence and total mass one for $|q| < 1$, and establishes the independent-copy affine refinement and law reflection. For $0 \leq q < 1$ it proves concentration on $[0, 1]$, including $q = 0$; at $q = 1/2$, `ProbabilityRepresentation.lean` identifies both the random series and its law with the classical dyadic construction.

This promotion does not prove that the topological support is exactly $[0, 1]$, nor does it define or analyze the general- q CDF, density, characteristic-function product, moments, cumulants, smoothness, or uniqueness of the affine fixed point. Those report claims remain frontier mathematics.

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1 Corpus audit and the novelty boundary

1.1 What was inspected

The requested tree contains four logically different kinds of material. First are the living synthesis documents: the primary Fabius–Rvachev exposition, the mathematical glossary, the Lean walkthrough, the non-elementarity article, the consolidated frontier volume, and the living Thue–Morse formula atlas. Second is a twelve-document Fourier-decay campaign containing proposed analyses, final syntheses, a gentle guide, and two comparative audits. Third is a seven-source paper library, including both the original and corrected TeX of one OCR conversion. Fourth is a frozen archive of thirty-two historical versions organized by provenance and topic. The resulting count is

$$6 + 12 + 7 + 32 = 57.$$

The exact paths are reproduced in [Appendix A](#) and in the companion plain-text ledger. The repository’s own archive README states that byte-identical historical copies were pruned and that the archive is grouped by purpose rather than serving as a naive mirror of every earlier path [22]. Thus the historical sources were treated as a novelty/provenance map, while the current synthesis documents were treated as the canonical statement of what the project already contains.

1.2 What the repository already does very well

The current primary exposition and frontier synthesis already cover, among much else,

- the probabilistic construction of the Fabius distribution and the Rvachev bump;
- exact dyadic evaluation, recurrences, bit-recursive algorithms, and denominator arithmetic;
- Gaussian-binomial and q -Pochhammer structures at the fixed base $q = 1/2$;
- Bernoulli cumulants, Bell-polynomial moments, Appell-type blocks, and exact coefficient extraction;
- geometric-node Lagrange identities and q -Richardson cancellation of geometric error modes;
- lower-Lambert endpoint variables, nonconstant periodic correction functions, and all-orders saddle machinery;
- the inverse Fabius function, totalization conventions, and dyadic inverse formulas;
- repeated-integration and box-spline approximants;
- extensive Thue–Morse identities, rarefied sums, Mahler equations, natural-boundary phenomena, and quantitative approximation questions;
- the quadratic logarithmic envelope, exceptional zeros, and several amplitude notions for the Fourier image of the up-function.

These features are visible in the primary exposition [18], the consolidated frontier volume [19], the living atlas [20], and the Fourier audit campaign [21]. The fixed-base q -calculus is especially important for the present report: the archive explicitly describes those documents as *base-1/2* articles [22]. Likewise, the archive records an existing “phase locking” mechanism for dyadic coefficient extraction. Accordingly, neither Gaussian-binomial notation nor phase locking by itself is claimed as new below.

1.3 Deliberate exclusions

Three tempting projects were rejected because they would duplicate the corpus too closely.

1. We do not offer a thirteenth derivation of the leading Fourier decay $\log |\widehat{\text{up}}(t)| \asymp -(\log |t|)^2 / (2 \log 2)$ with the known zero-set and limsup qualifications. The live Fourier drafts compare precisely such arguments.
2. We do not repackage the exact dyadic value web, the half-base Gaussian-binomial interpolation formula, or the existing geometric Richardson filter.
3. We do not relabel the existing dyadic lower-Lambert periodic asymptotic as a new “transseries.” The new phase-aware extrapolation theorem in [Section 9](#) applies to a different numerical geometry: inverse powers of a Lambert variable while its fractional phase is held fixed.

1.4 The transverse direction

The central observation is that the dyadic distribution is one member of a natural geometric one-parameter family. Making the ratio variable simultaneously exposes:

- a shape transition at the dyadic value $q = 1/2$;
- a majorization order in the parameter;
- a Gaussian limit as the geometric mesh becomes diffuse;
- a general Bernoulli–Bell calculus whose dyadic specialization recovers the repository’s formulas exactly;
- a change in Fourier-zero arithmetic from generic simple zeros to valuation-controlled multiplicities;
- a precise reason that the Thue–Morse block is special to base two.

This parameter deformation is the organizing principle of the report.

2 The geometric q -Fabius–Rvachev family

Throughout, fix

$$0 < q < 1, \quad c = 1 - q, \quad a = \log(1/q) > 0.$$

Let $(U_j)_{j \geq 0}$ be independent random variables, each uniform on $[0, 1]$.

Definition 2.1 (Geometric Fabius law). Define

$$X_q = c \sum_{j=0}^{\infty} q^j U_j, \quad Z_q = X_q - \frac{1}{2}, \quad Y_q = 2Z_q = 2X_q - 1. \quad (1)$$

Write F_q for the extended cumulative distribution function of X_q , f_q for its density, and u_q for the density of Y_q .

Because $0 \leq U_j \leq 1$ and $c \sum q^j = 1$, the series converges absolutely almost surely and $X_q \in [0, 1]$. Replacing every U_j by $1 - U_j$ shows

$$X_q \stackrel{d}{=} 1 - X_q, \quad F_q(x) = 1 - F_q(1 - x), \quad f_q(x) = f_q(1 - x). \quad (2)$$

The support is all of $[0, 1]$: every open subinterval of $[0, 1]$ receives positive probability by constraining finitely many U_j and leaving a sufficiently short tail free.

Current formalization boundary. The declarations `hasSum_geometricUniformWeight`, `geometricUniformSeries_mem_Icc`, and `geometricUniformDistribution_Icc` formalize the series, its unit total weight, and concentration on $[0, 1]$. They do not yet formalize the stronger assertion in the preceding sentence that every open subinterval has positive mass, so exact topological support remains open. Likewise `geometricUniformDistribution_reflection` proves $X_q \stackrel{d}{=} 1 - X_q$, but the displayed CDF and density symmetries still require the separate no-atoms and density layers.

Exact public API crosswalk. All names in this paragraph lie in the namespace `Fabius.ProbabilityRepresentation`. The generic almost-everywhere recursion assumes measurability of the tail statistic and combining map but not of the recursively described function; its pointwise wrapper is separate. Continuous-linear naturality requires completeness only in the source, pointwise scalar naturality needs neither completeness nor summability, and law-level identities retain the Borel and norm-summability hypotheses. The eleven generic declarations are `uniformProduct_map_of_ae_head_tail_recursion`, `uniformProduct_map_of_head_tail_recursion`, `weightedUniformSeries_map`, `weightedUniformSeries_smul`, `weightedUniformDistribution_map`, `weightedUniformDistribution_smul`, `weightedUniformDistribution_map_eq_shift`, `weightedUniformSeries_split_of_shift_eq_map`, `weightedUniformDistribution_split_of_shift_eq_map`, `weightedUniformSeries_geometric_split`, and `weightedUniformDistribution_geometric_split`. The dyadic-to-generic law bridge is `weightedSumDistribution_eq_weightedUniformDistribution`.

The compatibility surface consists of `weightedUniformSeries_smul_weights`, `uniformProduct_map_head_tail_function`, `weightedUniformDistribution_isProbabilityMeasure`, `weightedUniformDistribution_smul_weights`, `uniformProduct_map_head_tail_weightedUniformSeries`, `weightedUniformDistribution_unitInterval`, and `weightedUniformDistribution_compl_unitInterval`. In particular, `weightedUniformDistribution_smul_weights` is oriented as $\mu_{cw} = (x \mapsto cx) * \mu_w$.

The nineteen declarations in the named specialization are `geometricUniformWeight`, `geometricUniformWeight_zero`, `geometricUniformWeight_succ`, `hasSum_geometricUniformWeight`, `summable_norm_geometricUniformWeight`, `geometricUniformWeight_nonneg`, `geometricUniformSeries`, `continuous_geometricUniformSeries`, `measurable_geometricUniformSeries`, `geometricUniformSeries_split`, `geometricUniformSeries_reflect`, `geometricUniformSeries_mem_Icc`, `geometricUniformDistribution`, `geometricUniformDistribution_isProbabilityMeasure`, `uniformProduct_map_head_tail_geometricUniformSeries`, `geometricUniformDistribution_selfSimilar`, `geometricUniformDistribution_reflection`, `geometricUniformDistribution_Icc`, and `geometricUniformDistribution_compl_Icc`. Finally, `weightedCoordinateSum_eq_geometricUniformSeries_one_half` and `weightedSumDistribution_eq_geometricUniformDistribution_one_half` are the exact series- and law-level bridges at $q = 1/2$.

2.1 Recovery of the classical functions

At $q = 1/2$,

$$X_{1/2} = \sum_{j \geq 0} 2^{-j-1} U_j.$$

This is the standard bounded Fabius distribution. Moreover, if $V_j = 2U_j - 1 \sim \text{Unif}[-1, 1]$, then

$$Y_{1/2} = \sum_{j \geq 0} 2^{-j-1} V_j, \quad \widehat{u}_{1/2}(t) = \prod_{j \geq 1} \text{sinc}(t/2^j),$$

which is Rvachev's normalized up-function. The affine relation between densities is

$$u_q(y) = \frac{1}{2} f_q\left(\frac{y+1}{2}\right), \quad f_q(x) = 2u_q(2x-1). \quad (3)$$

Thus the free parameter deforms the Fabius CDF and the Rvachev bump simultaneously.

2.2 Distributional refinement

Let X'_q be an independent copy of X_q . Splitting off the first summand gives the basic self-similarity.

Proposition 2.2 (Affine refinement law). *One has*

$$X_q \stackrel{d}{=} cU_0 + qX'_q. \quad (4)$$

With the convention $F_q(x) = 0$ for $x \leq 0$ and $F_q(x) = 1$ for $x \geq 1$,

$$F_q(x) = \int_0^1 F_q\left(\frac{x-cu}{q}\right) du \quad (5)$$

$$= \frac{q}{c} \int_{(x-c)/q}^{x/q} F_q(v) dv, \quad (6)$$

$$f_q(x) = \frac{F_q(x/q) - F_q((x-c)/q)}{c}. \quad (7)$$

Proof. Equation (4) is the index shift

$$c \sum_{j \geq 0} q^j U_j = cU_0 + q \left(c \sum_{j \geq 0} q^j U_{j+1} \right).$$

Conditioning on U_0 gives (5). The substitution $v = (x-cu)/q$ yields (6), and differentiation gives (7); smoothness needed for the differentiation follows from [Theorem 3.1](#) below, or one may first interpret the derivative distributionally. $\square$

Current formalization boundary. The measure identity (4) is formalized exactly as geometricUniformDistribution_selfSimilar; its proof uses the public joint head–tail law uniformProduct_map_head_tail_geometricUniformSeries. The two CDF integrals and the density identity in this proposition have not yet been promoted, and the formal self-similarity theorem does not assert uniqueness of the affine measure fixed point.

For $q = 1/2$ and $0 \leq x \leq 1/2$, the second term in (7) vanishes, so

$$f_{1/2}(x) = 2F_{1/2}(2x),$$

the classical Fabius differential equation in bounded-CDF normalization.

Repeated differentiation gives a useful hierarchy.

Corollary 2.3 (Derivative refinement). *For every integer $r \geq 1$,*

$$f_q^{(r)}(x) = \frac{f_q^{(r-1)}(x/q) - f_q^{(r-1)}((x-c)/q)}{cq^r}. \quad (8)$$

In particular, every derivative vanishes at both endpoints of the support.

3 Infinite sinc products and smooth compact support

The centered uniform variable $U - 1/2$ has characteristic function $\text{sinc}(t/2)$. Independence therefore gives an infinite product.

Theorem 3.1 (Fourier product and rapid polynomial decay). *The characteristic functions of Z_q and Y_q are*

$$\phi_q(t) := \mathbb{E}e^{itZ_q} = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{cq^j t}{2}\right), \quad (9)$$

$$\widehat{u}_q(t) = \mathbb{E}e^{itY_q} = \prod_{j=0}^{\infty} \text{sinc}(cq^j t). \quad (10)$$

For every integer $N \geq 1$ and every $t \neq 0$,

$$|\phi_q(t)| \leq \left(\frac{2}{c}\right)^N q^{-N(N-1)/2} |t|^{-N}. \quad (11)$$

Consequently X_q and Y_q possess compactly supported C^∞ densities, and all of their derivatives are obtained by absolutely convergent Fourier inversion.

Proof. The product formulas follow from independence and the identity $\mathbb{E}e^{is(U-1/2)} = \text{sinc}(s/2)$. The product converges locally uniformly because $1 - \text{sinc} z = \mathcal{O}(z^2)$ and $\sum q^{2j} < \infty$. For the first N factors, use $|\text{sinc} x| \leq \min(1, |x|^{-1})$; bound all remaining factors by one. This gives

$$|\phi_q(t)| \leq \prod_{j=0}^{N-1} \frac{2}{cq^j |t|} = \left(\frac{2}{c}\right)^N q^{-N(N-1)/2} |t|^{-N}.$$

Given $r \geq 0$, choose $N > r + 1$. Then $t^r \phi_q(t) \in L^1(\mathbb{R})$, so Fourier inversion gives a continuous r th derivative. Since r is arbitrary, the density is C^∞ . Compact support was already established from the random series. $\square$

Remark 3.2 (Relation to the existing Fourier-decay campaign). Bound (11) is intentionally elementary. Optimizing N as a function of $|t|$ recovers a quadratic logarithmic envelope, but the dyadic case has already been analyzed in much finer arithmetic and phase detail in the repository. The genuinely new content here is the q -deformation and the zero arithmetic in [Section 8](#), not a claim of a new sharp dyadic envelope.

4 Bernoulli cumulants, Bell moments, and q -binomial scale geometry

4.1 Exact cumulants

Let B_n denote the Bernoulli numbers in the convention

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} B_n \frac{t^n}{n!}, \quad B_1 = -\frac{1}{2}.$$

For a uniform random variable U on $[0, 1]$, the cumulants are $\kappa_1(U) = 1/2$ and $\kappa_n(U) = B_n/n$ for $n \geq 2$.

Theorem 4.1 (Bernoulli cumulants). *For the centered law $Z_q = X_q - 1/2$,*

$$\kappa_{2m+1}(Z_q) = 0 \quad (m \geq 0), \quad (12)$$

where the case $m = 0$ means $\kappa_1 = 0$, and

$$\boxed{\kappa_{2m}(Z_q) = \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}}} \quad (m \geq 1). \quad (13)$$

In particular,

$$\sigma_q^2 := \text{Var}(X_q) = \frac{1-q}{12(1+q)}. \quad (14)$$

At $q = 1/2$,

$$\kappa_{2m}(Z_{1/2}) = \frac{B_{2m}}{2m(2^{2m} - 1)}, \quad (15)$$

exactly the dyadic Bernoulli-cumulant formula already present in the repository.

Proof. Cumulants add over independent sums and scale homogeneously. Hence for $n \geq 2$,

$$\kappa_n(Z_q) = \sum_{j \geq 0} (cq^j)^n \kappa_n(U_j - 1/2) = \frac{B_n}{n} \frac{c^n}{1-q^n}.$$

All odd Bernoulli numbers beyond B_1 vanish, while symmetry gives the vanishing first cumulant. Substituting $n = 2$ yields (14); substituting $q = 1/2$ gives (15). $\square$

4.2 Bell-polynomial moments and a triangular recurrence

Let $\mathcal{B}_n(x_1, \dots, x_n)$ denote the complete exponential Bell polynomial, characterized by

$$\exp\left(\sum_{r \geq 1} x_r \frac{t^r}{r!}\right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}.$$

Corollary 4.2 (Exact moments). *For every $n \geq 0$,*

$$\mathbb{E}Z_q^n = \mathcal{B}_n(0, \kappa_2(Z_q), \dots, \kappa_n(Z_q)), \quad (16)$$

$$\mathbb{E}X_q^n = \mathcal{B}_n\left(\frac{1}{2}, \kappa_2(Z_q), \dots, \kappa_n(Z_q)\right). \quad (17)$$

The raw moments $m_n(q) = \mathbb{E}X_q^n$ also satisfy

$$m_0(q) = 1, \quad m_n(q) = \frac{1}{1-q^n} \sum_{k=1}^n \binom{n}{k} \frac{c^k q^{n-k}}{k+1} m_{n-k}(q). \quad (18)$$

Proof. The Bell formulas are the moment-cumulant relation. For the recurrence, raise $cU_0 + qX'_q$ to the n th power, take expectations, use $\mathbb{E}U_0^k = 1/(k+1)$, and move the $k = 0$ term $q^n m_n(q)$ to the left. $\square$

The first centered even moments are therefore

$$\begin{aligned} \mathbb{E}Z_q^2 &= \kappa_2, \\ \mathbb{E}Z_q^4 &= 3\kappa_2^2 + \kappa_4, \\ \mathbb{E}Z_q^6 &= 15\kappa_2^3 + 15\kappa_2\kappa_4 + \kappa_6, \end{aligned}$$

with every cumulant given by (13). This is an explicit all-orders moment engine involving only Bernoulli numbers, rational operations, and Bell contraction.

4.3 A Bernoulli–Appell inverse to geometric averaging

Let

$$M_q(t) = \mathbb{E}e^{tX_q} = \prod_{j \geq 0} \frac{e^{cq^j t} - 1}{cq^j t}. \quad (19)$$

Define an Appell sequence $(P_{q,n})_{n \geq 0}$ by

$$\frac{e^{xt}}{M_q(t)} = \sum_{n \geq 0} P_{q,n}(x) \frac{t^n}{n!}. \quad (20)$$

Theorem 4.3 (Geometric Bernoulli–Appell polynomials). *The polynomials $P_{q,n}$ satisfy*

$$\frac{d}{dx} P_{q,n}(x) = n P_{q,n-1}(x), \quad (21)$$

$$\mathbb{E}[P_{q,n}(x + X_q)] = x^n, \quad (22)$$

$$P_{q,n}(x) = \mathcal{B}_n\left(x - \frac{1}{2}, -\kappa_2(Z_q), \dots, -\kappa_n(Z_q)\right). \quad (23)$$

They obey the refinement recurrence

$$P_{q,n}(x) = \sum_{k=0}^n \binom{n}{k} B_k c^k q^{n-k} P_{q,n-k}\left(\frac{x}{q}\right). \quad (24)$$

Proof. Differentiate (20) in x to obtain (21). Multiplying by $M_q(t)$ and taking expectation after replacing x by $x + X_q$ gives e^{xt} , proving (22). Taking the logarithm of the left side of (20) gives

$$\left(x - \frac{1}{2}\right)t - \sum_{r \geq 2} \kappa_r(Z_q) \frac{t^r}{r!},$$

which is exactly the Bell generating function in (23).

Finally, the distributional refinement implies

$$M_q(t) = \frac{e^{ct} - 1}{ct} M_q(qt).$$

Therefore

$$\frac{e^{xt}}{M_q(t)} = \frac{ct}{e^{ct} - 1} \frac{e^{(x/q)(qt)}}{M_q(qt)}.$$

Expand the first factor as $\sum B_k c^k t^k / k!$ and the second as $\sum P_{q,r}(x/q) q^r t^r / r!$; coefficient comparison yields (24). $\square$

The operator meaning is transparent. For $c > 0$, define uniform averaging

$$(\mathcal{A}_c p)(x) = \frac{1}{c} \int_0^c p(x+u) du = \frac{e^{cD} - 1}{cD} p(x), \quad D = \frac{d}{dx}.$$

Then

$$\mathcal{A}_c^{-1} x^n = c^n B_n(x/c), \quad (25)$$

where $B_n(x)$ is the Bernoulli polynomial. The sequence $P_{q,n}$ is obtained by composing these Bernoulli inverses over all geometric scales $c, cq, cq^2, \dots$. This gives a direct operator bridge between Bernoulli polynomials, the random series, and the Appell blocks used in exact dyadic arithmetic.

4.4 Finite-stage Gaussian-binomial geometry

Let $w_j = cq^j$ for $j = 0, \dots, N-1$. The elementary symmetric functions of these scales are Gaussian-binomial coefficients.

Proposition 4.4 (*q-binomial scale identity*). *For every integer $N \geq 0$,*

$$\prod_{j=0}^{N-1} (1 + zw_j) = \sum_{k=0}^N c^k q^{k(k-1)/2} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k. \quad (26)$$

Equivalently,

$$e_k(w_0, \dots, w_{N-1}) = c^k q^{k(k-1)/2} \begin{bmatrix} N \\ k \end{bmatrix}_q. \quad (27)$$

Proof. This is the finite q -binomial theorem applied to $\prod_{j=0}^{N-1} (1 + cq^j z)$. $\square$

Remark 4.5 (What is and is not new here). The repository already exploits the same Gaussian-binomial algebra at the fixed dyadic base $q = 1/2$ for coefficient extraction and exact Fabius values. The new point is that the same q now parametrizes an actual family of probability laws. Thus differentiation in q , limits $q \uparrow 1$, shape transitions, and arithmetic changes of the zero set become meaningful analytic questions rather than formal changes of notation.

5 Parameter geometry: majorization, shape, and concentration

5.1 A convex-order flow

Write the decreasing weight sequence

$$\mathbf{w}(q) = (c, cq, cq^2, \dots).$$

Its partial sums are

$$\sum_{j=0}^{n-1} w_j(q) = 1 - q^n.$$

Hence, if $0 < q_1 < q_2 < 1$, then every partial sum of $\mathbf{w}(q_1)$ is larger while both total sums equal one. In the language of infinite majorization, $\mathbf{w}(q_1) \succ \mathbf{w}(q_2)$.

Theorem 5.1 (Monotone contraction in convex order). *If $0 < q_1 < q_2 < 1$, then*

$$X_{q_2} \preceq_{\text{cx}} X_{q_1}; \quad (28)$$

that is, for every convex function $\Phi : [0, 1] \rightarrow \mathbb{R}$,

$$\mathbb{E}\Phi(X_{q_2}) \leq \mathbb{E}\Phi(X_{q_1}). \quad (29)$$

Consequently, for every $p \geq 1$,

$$\mathbb{E}|X_{q_2} - \tfrac{1}{2}|^p \leq \mathbb{E}|X_{q_1} - \tfrac{1}{2}|^p. \quad (30)$$

Proof. For finitely many i.i.d. integrable random variables, the map

$$(a_1, \dots, a_N) \mapsto \mathbb{E}\Phi\left(\sum_{j=1}^N a_j U_j\right)$$

is symmetric and convex in the coefficient vector, hence Schur-convex. Therefore it increases under majorization. The same statement extends to nonnegative ℓ^1 coefficient sequences by truncation:

the omitted weighted tail is bounded by the omitted ℓ^1 mass, so the corresponding sums converge uniformly, and every convex function on the compact interval $[0, 1]$ is continuous. Since

$$\sum_{j=0}^{n-1} w_j(q_1) = 1 - q_1^n \geq 1 - q_2^n = \sum_{j=0}^{n-1} w_j(q_2)$$

for every n , while both full sums are one, infinite majorization gives (29). Take $\Phi(x) = |x - 1/2|^p$ for (30). $\square$

This theorem turns q into a genuine concentration parameter. At $q \downarrow 0$, the law approaches a single uniform random variable. As $q \uparrow 1$, its weight vector is spread among increasingly many small summands, and the law contracts toward $1/2$.

5.2 A uniform sub-Gaussian bound

Hoeffding's lemma applied to each centered term gives

$$\log \mathbb{E} e^{tZ_q} \leq \frac{t^2}{8} \sum_{j \geq 0} c^2 q^{2j} = \frac{t^2}{8} \frac{1-q}{1+q}.$$

Corollary 5.2 (Concentration). *For every $r > 0$,*

$$\mathbb{P}(|X_q - \tfrac{1}{2}| \geq r) \leq 2 \exp\left(-\frac{2(1+q)}{1-q} r^2\right). \quad (31)$$

The exponent has the correct $1/(1-q)$ concentration scale as $q \uparrow 1$, though it is not intended to replace the exact Gaussian limit below.

5.3 Log-concavity and an exact plateau transition

Every finite partial sum of (1) has a density that is a convolution of uniform interval densities, hence is log-concave. Passing to the locally uniform density limit shows that f_q is log-concave on $(0, 1)$. Symmetry then makes $1/2$ a global maximizer. The first scale competes with the total length of all remaining scales:

$$w_0 = c, \quad \sum_{j \geq 1} w_j = q.$$

Equality occurs exactly at $q = 1/2$.

Theorem 5.3 (Shape phase transition). *For every $0 < q < 1$, the density f_q is symmetric and log-concave.*

(i) *If $0 < q < 1/2$, then the exact maximal plateau is*

$$f_q(x) = \frac{1}{1-q} \iff q \leq x \leq 1-q. \quad (32)$$

Outside that interval, $f_q(x) < 1/(1-q)$.

(ii) *If $1/2 \leq q < 1$, then f_q has no nontrivial plateau. It is strictly increasing on $(0, 1/2)$ and strictly decreasing on $(1/2, 1)$, so $1/2$ is its unique maximizer.*

Proof. Log-concavity follows from closure of log-concave densities under convolution and locally uniform limits. If $q < 1/2$ and $q \leq x \leq c$, then $x/q \geq 1$ and $(x-c)/q \leq 0$, so (7) gives $f_q(x) = 1/c$. If $0 < x < q$, then $F_q(x/q) < 1$ and the second CDF term is zero, so $f_q(x) < 1/c$; symmetry handles $x > c$. This proves (i).

For (ii), first take $q = 1/2$. On $(0, 1/2)$ the dyadic refinement reads $f_{1/2}(x) = 2F_{1/2}(2x)$. The support statement above makes $F_{1/2}$ strictly increasing, so $f_{1/2}$ is strictly increasing there; symmetry handles the right half.

Now assume $q > 1/2$ and work in centered coordinates. Suppose the density has a nontrivial central plateau of half-width $h > 0$. In the decomposition

$$Z_q = c(U_0 - \tfrac{1}{2}) + qZ'_q,$$

let g be the density of qZ'_q , so $g(z) = q^{-1}f_Z(z/q)$ and its central plateau has half-width qh . Convolution with the uniform density gives

$$f'_Z(y) = \frac{g(y + c/2) - g(y - c/2)}{c}. \quad (33)$$

For $0 < y < \min\{h, (q - c)/2\}$, both points lie in the interior of the support of g . Since $f'_Z(y) = 0$, symmetry and log-concavity force g to be constant from the origin out to $c/2 + y$: a symmetric log-concave density cannot have an off-center constant interval without also being constant between that interval and the origin. Thus $qh \geq c/2 + y$ throughout this range. If $h \leq (q - c)/2$, let $y \uparrow h$ to obtain $qh \geq c/2 + h$, impossible because $q < 1$. If $h > (q - c)/2$, let $y \uparrow (q - c)/2$ to obtain $qh \geq q/2$, hence $h \geq 1/2$, which would make the density constant on its entire support, also impossible. Therefore no plateau exists.

Finally, a symmetric log-concave density is nonincreasing to the right of its center. If it took the same value at two distinct positive distances from the center, concavity of its logarithm would force a central plateau. Hence it decreases strictly, and symmetry gives the left-hand statement. $\square$

Conjecture 5.4 (Strict log-concavity beyond the transition). *For every $q > 1/2$, $\log f_q$ is strictly concave on $(0, 1)$.*

The theorem proves strict unimodality but not strict concavity of the logarithm. A plausible route is to quantify the strictness in the Prékopa–Leindler step for an infinite convolution whose largest interval is shorter than the sum of the others.

6 Gaussian degeneration and an Edgeworth frontier

Let

$$G_q = \frac{Z_q}{\sigma_q}, \quad \sigma_q^2 = \frac{1 - q}{12(1 + q)}.$$

The largest normalized summand has magnitude of order $\sqrt{1 - q}$, while the total variance is one. This is a triangular-array central-limit regime.

Theorem 6.1 (Gaussian limit). *As $q \uparrow 1$,*

$$G_q \xrightarrow{d} \mathcal{N}(0, 1). \quad (34)$$

The standardized even cumulants are

$$\gamma_{2m}(q) := \kappa_{2m}(G_q) = \frac{B_{2m}}{2m} 12^m \frac{(1 - q^2)^m}{1 - q^{2m}}. \quad (35)$$

In particular,

$$\gamma_4(q) = -\frac{6}{5} \frac{1 - q^2}{1 + q^2}, \quad (36)$$

$$\gamma_6(q) = \frac{48}{7} \frac{(1 - q^2)^2}{1 + q^2 + q^4}. \quad (37)$$

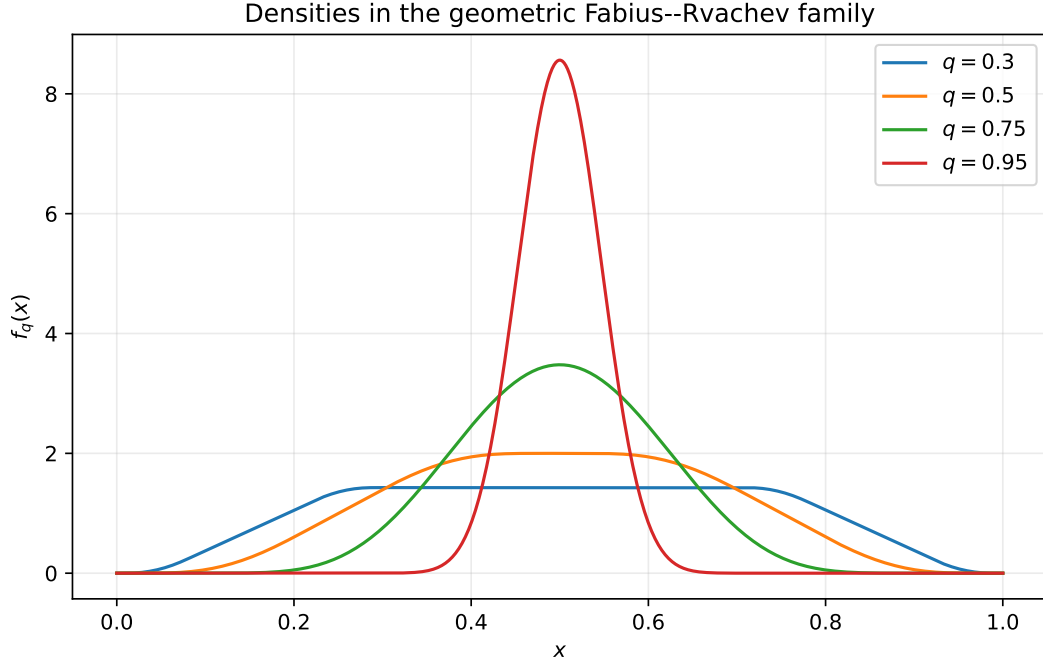


Figure 1: Deterministic fixed-point approximations of f_q . The exact plateau for $q = 0.30$ ends at $x = q$ and $x = 1 - q$. At $q = 1/2$ the plateau collapses to a point; for larger q the law becomes sharply concentrated and strictly unimodal.

For every fixed $T < \infty$, uniformly for $|t| \leq T$,

$$\log \mathbb{E} e^{itG_q} = -\frac{t^2}{2} + \frac{\gamma_4(q)t^4}{4!} + \mathcal{O}_T((1-q)^2). \quad (38)$$

Proof. Write

$$G_q = \sum_{j \geq 0} \alpha_{q,j} (U_j - \tfrac{1}{2}), \quad \alpha_{q,j} = \frac{cq^j}{\sigma_q}.$$

The summands are bounded, their variances sum to one, and

$$\max_j |\alpha_{q,j}| = \frac{c}{\sigma_q} = \sqrt{12(1-q)(1+q)} \longrightarrow 0.$$

The Lindeberg condition is therefore immediate, proving (34). Dividing (13) by σ_q^{2m} gives (35); elementary factorization gives (36)–(37).

For the compact-frequency expansion, expand the logarithm of each sinc factor. The maximum factor argument is $\mathcal{O}_T(\sqrt{1-q})$, so the expansion is uniform. The quadratic terms sum to $-t^2/2$, the quartic terms sum to $\gamma_4 t^4/4!$, and the sum of all terms of order six and higher is $\mathcal{O}_T((1-q)^2)$ because $\gamma_{2m}(q) = \mathcal{O}((1-q)^{m-1})$ for $m \geq 3$. $\square$

Formally inverting (38) suggests

$$\sigma_q f_q\left(\frac{1}{2} + \sigma_q x\right) = \varphi(x) \left[1 + \frac{\gamma_4(q)}{24} \text{He}_4(x)\right] + \mathcal{O}((1-q)^2), \quad (39)$$

where $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and $\text{He}_4(x) = x^4 - 6x^2 + 3$. At the center this predicts

$$f_q(1/2) = \frac{1}{\sqrt{2\pi}\sigma_q} \left[1 + \frac{\gamma_4(q)}{8} + \mathcal{O}((1-q)^2)\right]. \quad (40)$$

For $q = 0.95$, the deterministic computation gives $f_q(1/2) = 8.56266675$, while the Gaussian term with the quartic correction gives 8.56 to the displayed precision.

Conjecture 6.2 (Uniform density Edgeworth expansion). *For every fixed integer $r \geq 1$, the normalized density of G_q admits a uniform all-orders Edgeworth expansion through order $(1 - q)^r$, with coefficients obtained by Bell contraction of the explicit standardized cumulants (35). In particular, (39) holds uniformly in $x \in \mathbb{R}$ with error $\mathcal{O}((1 - q)^2)$ after inclusion of the complete order- $(1 - q)^2$ Hermite polynomial.*

The missing ingredient is not the coefficient algebra, which is explicit, but a clean q -uniform Fourier-tail estimate strong enough to justify termwise inversion on the whole line.

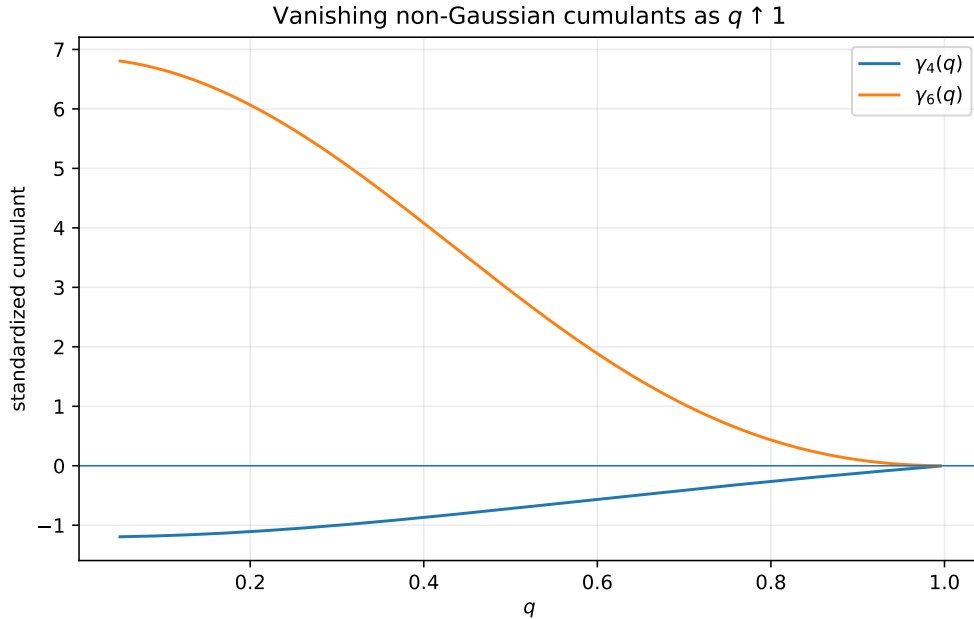


Figure 2: The exact standardized cumulants $\gamma_4(q)$ and $\gamma_6(q)$. Their decay to zero quantifies the Gaussian degeneration as $q \uparrow 1$.

7 A rigorous endpoint law and Lambert- W inversion

The exact dyadic repository contains a much sharper lower-Lambert expansion with periodic corrections. The purpose here is different: establish, for every q , the first two universal logarithmic scales by a short probability argument, then invert them rigorously. Set

$$L = \log(1/x), \quad a = \log(1/q).$$

7.1 Finite-simplex upper bound

For $n \geq 1$, let

$$S_n = c \sum_{j=0}^{n-1} q^j U_j.$$

If $0 < x \leq cq^{n-1}$, then the simplex $\sum_{j < n} cq^j u_j \leq x$ lies entirely inside the unit cube. Its volume is exact:

$$\mathbb{P}(S_n \leq x) = \frac{x^n}{n! c^n q^{n(n-1)/2}}. \quad (41)$$

Since $X_q \geq S_n$, this is an upper bound for $F_q(x)$.

7.2 A matching product lower bound

Choose n so that the deterministic tail is at most $x/2$:

$$c \sum_{j=n}^{\infty} q^j = q^n \leq x/2.$$

If, for every $0 \leq j < n$,

$$U_j \leq \frac{x}{2ncq^j},$$

then the first n summands contribute at most $x/2$, and hence $X_q \leq x$. For small x the displayed thresholds are all at most one, giving the lower bound

$$F_q(x) \geq \frac{x^n}{(2nc)^n q^{n(n-1)/2}}. \quad (42)$$

Theorem 7.1 (Two-term logarithmic endpoint law). *For each fixed $q \in (0, 1)$, as $x \downarrow 0$,*

$$\log F_q(x) = -\frac{L^2}{2a} - \frac{L \log L}{a} + \mathcal{O}_q(L), \quad L = \log(1/x), \quad a = \log(1/q). \quad (43)$$

By symmetry, the same law holds for $1 - F_q(1 - x)$.

Proof. For the upper bound, choose

$$n_+(x) = \left\lfloor \frac{L + \log c}{a} \right\rfloor + 1.$$

Then $x \leq cq^{n_+ - 1}$, so (41) applies. Taking logarithms gives

$$\log F_q(x) \leq -n_+ L - \log(n_+!) - n_+ \log c + \frac{a}{2} n_+ (n_+ - 1). \quad (44)$$

Since $n_+ = L/a + \mathcal{O}_q(1)$, Stirling's formula yields

$$-n_+ L + \frac{a}{2} n_+^2 = -\frac{L^2}{2a} + \mathcal{O}_q(L), \quad -\log(n_+!) = -\frac{L \log L}{a} + \mathcal{O}_q(L).$$

Thus the right side of (44) is

$$-\frac{L^2}{2a} - \frac{L \log L}{a} + \mathcal{O}_q(L).$$

For the lower bound, take

$$n_-(x) = \left\lceil \frac{\log(2/x)}{a} \right\rceil.$$

Then $q^{n_-} \leq x/2$. The ceiling relation also gives $q^{n_- - 1} > x/2$, so the largest threshold in the product event is at most $1/(n_- c)$ and is therefore below one for sufficiently small x . Apply (42) and take logarithms:

$$\log F_q(x) \geq -n_- L - n_- \log(2nc) + \frac{a}{2} n_- (n_- - 1). \quad (45)$$

Again $n_- = L/a + \mathcal{O}_q(1)$, so the quadratic terms contribute $-L^2/(2a) + \mathcal{O}_q(L)$ and the $-n_- \log n_-$ term contributes $-(L \log L)/a + \mathcal{O}_q(L)$. The upper and lower bounds match at the two displayed scales. $\square$

Remark 7.2 (Why the second term is already nontrivial). The quadratic term comes from balancing the small probability of constraining roughly L/a coordinates against the geometric Jacobian $q^{n(n-1)/2}$. The additional $-L \log L/a$ term is the factorial entropy of the finite simplex (upper bound) and the n^{-n} cost of a balanced coordinate allocation (lower bound). Thus the second scale is not an artifact of saddle notation; it is forced by elementary high-dimensional geometry.

7.3 Inverse endpoint law

Let $Q_q(y) = F_q^{-1}(y)$ for $0 < y < 1$, and put

$$Y = \log(1/y), \quad L_y = \log \frac{1}{Q_q(y)}, \quad S = \sqrt{2aY}.$$

Equation (43) is equivalent to

$$2aY = L_y^2 + 2L_y \log L_y + \mathcal{O}_q(L_y). \quad (46)$$

Since $(\log L_y)^2 = o(L_y)$, one obtains

$$S = L_y + \log L_y + \mathcal{O}_q(1).$$

The inverse of $r \mapsto r + \log r$ is $r = W(e^s)$.

Corollary 7.3 (Lambert- W quantile law). *As $y \downarrow 0$,*

$$\boxed{\log \frac{1}{Q_q(y)} = W\left(\exp \sqrt{2a \log(1/y)}\right) + \mathcal{O}_q(1)} \quad (47)$$

$$= \sqrt{2a \log(1/y)} - \frac{1}{2} \log(2a \log(1/y)) + \mathcal{O}_q(1). \quad (48)$$

The corresponding upper-end quantile follows from symmetry: $Q_q(1-y) = 1 - Q_q(y)$.

Proof. From (46),

$$S^2 = (L_y + \log L_y)^2 + \mathcal{O}_q(L_y),$$

so $S = L_y + \log L_y + \mathcal{O}_q(1)$. If $r + \log r = s$, then $re^r = e^s$ and $r = W(e^s)$. Since the derivative of $r + \log r$ is bounded away from zero for large r , the additive $\mathcal{O}_q(1)$ perturbation in s changes r by $\mathcal{O}_q(1)$. This proves (47). The standard large-argument expansion $W(e^S) = S - \log S + \mathcal{O}((\log S)/S)$ gives (48). $\square$

This is deliberately weaker than the repository's all-orders dyadic lower-Lambert machinery, but it is rigorous for every q and directly links the inverse Fabius scale to Lambert W .

7.4 The expected periodic completion

The Laplace transform is

$$\mathcal{L}_q(s) = \mathbb{E}e^{-sX_q} = \prod_{j=0}^{\infty} \frac{1 - e^{-csq^j}}{csq^j}. \quad (49)$$

It is a harmonic sum over the logarithmic lattice $\log s - ja$. Mellin analysis of such sums normally produces a polynomial in the logarithmic scale plus a periodic function of its fractional part [14]. This mechanism is exactly what occurs in the dyadic endpoint expansion already present in the repository.

Conjecture 7.4 (Full geometric endpoint transseries). *For every fixed $q \in (0, 1)$ there exists a natural lower-Lambert saddle coordinate $\lambda_q(x) \rightarrow \infty$ and smooth 1-periodic functions $A_{q,r} : \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}$ such that*

$$F_q(x) = \exp(-S_q(\lambda_q(x))) \lambda_q(x)^{-1/2} \left[A_{q,0}(\{\lambda_q(x)\}) + \sum_{r=1}^N \frac{A_{q,r}(\{\lambda_q(x)\})}{\lambda_q(x)^r} + \mathcal{O}_{q,N}(\lambda_q^{-N-1}) \right], \quad (50)$$

where S_q has an explicit expansion whose first two consequences are (43). The leading periodic amplitude $A_{q,0}$ is positive and nonconstant for generic q .

Problem 7.5 (Fourier modes of the endpoint phase). *Determine the Mellin-residue formula for the Fourier coefficients of $A_{q,0}$ and classify those q for which one or more nonzero modes vanish. The dyadic development suggests nonvanishing, but the free parameter may admit exceptional arithmetic ratios.*

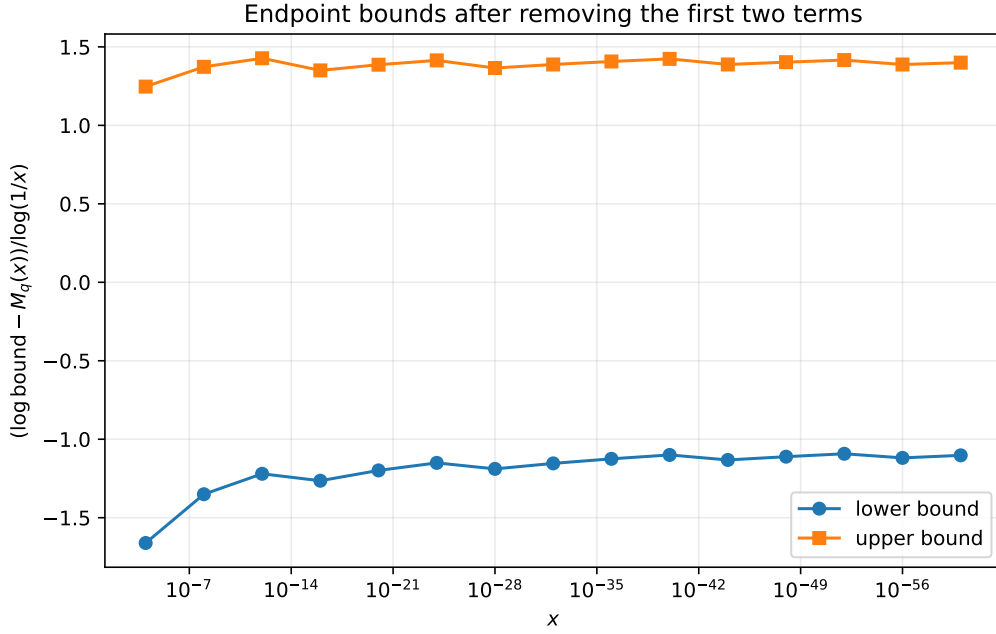


Figure 3: For $q = 1/2$, the rigorous upper and lower logarithmic bounds after subtracting the two universal terms and dividing by L . Bounded residuals are exactly what [Theorem 7.1](#) predicts.

8 Fourier-zero arithmetic and the special role of base two

The product (10) makes the zero set explicit:

$$\mathcal{Z}_q = \left\{ \frac{\pi k}{cq^j} : k \in \mathbb{Z} \setminus \{0\}, j \geq 0 \right\}. \quad (51)$$

Each individual factor has a simple nonzero zero. Multiplicity therefore counts collisions between geometric scales.

Theorem 8.1 (Reciprocal-integer multiplicities). *Let $q = 1/b$ with an integer $b \geq 2$. Then every nonzero zero of $\hat{u}_q$ lies on the lattice*

$$t = \frac{\pi N}{1 - 1/b}, \quad N \in \mathbb{Z} \setminus \{0\}. \quad (52)$$

Its multiplicity is

$$\text{mult}\left(\frac{\pi N}{1 - 1/b}\right) = 1 + v_b(|N|), \quad (53)$$

where $v_b(N) = \max\{r \geq 0 : b^r \mid N\}$ uses divisibility by the whole base b , which need not be prime.

Proof. At $q = 1/b$, the j th factor is $\text{sinc}(cb^{-j}t)$. At $t = \pi N/c$, it vanishes exactly when $b^j \mid N$. Thus the vanishing factors are indexed by $j = 0, 1, \dots, v_b(|N|)$, and their simple multiplicities add. Conversely, every zero of the j th factor has the form $t = \pi kb^j/c = \pi N/c$ for an integer $N \neq 0$. $\square$

For the classical up-function $b = 2$, the zeros are at $2\pi N$ in the convention (10), with multiplicity $1 + v_2(N)$.

Proposition 8.2 (Generic simplicity criterion). *If $q^d \notin \mathbb{Q}$ for every integer $d \geq 1$, then every nonzero zero of $\hat{u}_q$ is simple.*

Proof. A collision between scales $j < k$ would give

$$\frac{k_1}{q^j} = \frac{k_2}{q^k} \implies q^{k-j} = \frac{k_2}{k_1} \in \mathbb{Q}$$

for nonzero integers k_1, k_2 , contrary to the hypothesis. $\square$

Problem 8.3 (Complete collision classification). *For algebraic q such that $q^d \in \mathbb{Q}$ for some d , describe the zero set as a union of arithmetic lattices with a closed multiplicity formula. The reciprocal-integer theorem is the cleanest case; rational $q = r/s$ and radical rational $q = (r/s)^{1/d}$ should lead to a finite-state valuation problem.*

8.1 Why the contiguous Thue–Morse block is uniquely dyadic

For an integer base $b \geq 2$ and $N \in \mathbb{N}$, consider the finite product

$$P_{b,N}(z) = \prod_{j=0}^{N-1} (1 - z^{b^j}). \quad (54)$$

Expanding by subsets gives

$$P_{b,N}(z) = \sum_{\varepsilon \in \{0,1\}^N} (-1)^{\varepsilon_0 + \dots + \varepsilon_{N-1}} z^{\varepsilon_0 + \varepsilon_1 b + \dots + \varepsilon_{N-1} b^{N-1}}. \quad (55)$$

Proposition 8.4 (Dyadic contiguity). *The exponents in (55) form a complete interval of consecutive integers if and only if $b = 2$ or $N \leq 1$. In the dyadic case,*

$$P_{2,N}(z) = \sum_{n=0}^{2^N-1} (-1)^{\text{wt}_2(n)} z^n, \quad (56)$$

which is the contiguous signed Thue–Morse block. For every $b \geq 3$ and $N \geq 2$, the exponents are precisely the integers whose first N base- b digits belong to $\{0, 1\}$; they form a sparse Cantor-type set with gaps.

Proof. For $b = 2$, binary expansion is a bijection between $\{0, 1\}^N$ and $\{0, 1, \dots, 2^N - 1\}$, and the subset size is the binary digit sum. For every base, the exponent set is $\{0\}$ when $N = 0$ and $\{0, 1\}$ when $N = 1$. Now suppose $b \geq 3$ and $N \geq 2$. The exponents 0, 1, and b occur, whereas 2 cannot occur: reduction modulo b would force its units digit to lie in $\{0, 1\}$. Hence the exponent set has a gap between 1 and b and is not an interval. The stated digit characterization is immediate from (55). $\square$

This gives a structural explanation for a feature that can otherwise look accidental. The reciprocal-integer Fourier products exist for every base b , and their zero multiplicities are controlled by v_b . But only $b = 2$ makes the subset-product coefficients occupy every integer index at every nontrivial length $N \geq 2$. The Thue–Morse sequence is therefore not merely one member of a uniform b -adic family; it is the unique persistent no-gap member.

Problem 8.5 (Sparse b -adic signed operators). *For $b \geq 3$, develop the analogue of the dyadic signed extension on the sparse digit set $\{0, 1\}$ in base b . Natural questions include transfer matrices for rarefied sums, operator spectra, exact cancellation order, and whether the missing digits can be supplied by multinomial factors to produce a different automatic sequence with a contiguous block.*

9 Phase-aware Lagrange and Richardson extrapolation

The dyadic corpus already contains two ingredients that should not be conflated: geometric-node Lagrange extraction and q -Richardson cancellation of errors such as 2^{-rn} . Endpoint saddle expansions introduce a different obstacle: coefficients depend periodically on the fractional part of a Lambert variable. Ordinary extrapolation across samples with different phases mixes different coefficient functions and can lose its nominal order. The cure is to move by integer increments in the Lambert variable.

9.1 A phase-preserving interpolation theorem

Let $u \in \mathbb{R}/\mathbb{Z}$ be fixed. Suppose an observable has the phase-class representation

$$R(\lambda) = L + h_u(1/\lambda), \quad \lambda \equiv u \pmod{1}, \quad (57)$$

where h_u is smooth near zero and $h_u(0) = 0$. Use the samples $\lambda_m = \lambda + m$, $m = 0, \dots, p$; all have the same fractional phase.

Theorem 9.1 (Phase-aware Lagrange filter). *Let $p \in \mathbb{N}$. Choose $\lambda_0 > 0$ sufficiently large that $h_u \in C^{p+1}[0, 1/\lambda_0]$. For every $\lambda \geq \lambda_0$ satisfying $\lambda \equiv u \pmod{1}$, and for $m = 0, \dots, p$, define*

$$w_m^{(p)}(\lambda) = (-1)^{p-m} \frac{(\lambda + m)^p}{m!(p-m)!}. \quad (58)$$

Then

$$\sum_{m=0}^p w_m^{(p)}(\lambda) = 1, \quad (59)$$

$$\sum_{m=0}^p \frac{w_m^{(p)}(\lambda)}{(\lambda + m)^r} = 0 \quad (1 \leq r \leq p), \quad (60)$$

and

$$\sum_{m=0}^p w_m^{(p)}(\lambda) R(\lambda + m) = L + \mathcal{O}_{u,p}(\lambda^{-p-1}) \quad (\lambda \rightarrow +\infty, \lambda \equiv u \pmod{1}). \quad (61)$$

More precisely, the error is bounded by

$$\frac{\|h_u^{(p+1)}\|_{L^\infty[0,1/\lambda]}}{(p+1)!} \prod_{m=0}^p \frac{1}{\lambda + m}. \quad (62)$$

Proof. Interpolate $h_u(t)$ at the nodes $t_m = (\lambda + m)^{-1}$ and evaluate the degree- p interpolant at $t = 0$. The Lagrange basis weight is

$$\ell_m(0) = \prod_{r \neq m} \frac{-t_r}{t_m - t_r}.$$

Since

$$t_m - t_r = \frac{r - m}{(\lambda + m)(\lambda + r)},$$

a direct cancellation gives

$$\ell_m(0) = (-1)^{p-m} \frac{(\lambda + m)^p}{m!(p-m)!},$$

which is (58). Exactness on $1, t, \dots, t^p$ gives (59)–(60). The standard interpolation remainder at zero is

$$\frac{h_u^{(p+1)}(\xi)}{(p+1)!} \prod_{m=0}^p (0 - t_m),$$

which proves (62) and hence (61). $\square$

Remark 9.2 (Conditioning). The weights grow individually like λ^p and cancel strongly. The theorem is exact, but a floating-point implementation should use extended precision, compensated summation, or a barycentric form. The deterministic experiment in [Section 10](#) uses moderate orders and directly verifies the predicted slope.

9.2 Fourier phase averaging

A complementary operation removes periodic modes rather than preserving a single phase.

Theorem 9.3 (Root-of-unity phase filter). *Let $M \in \mathbb{N}$ with $M \geq 1$, and let $\Psi(u) = \sum_{k \in \mathbb{Z}} c_k e^{2\pi i k u}$ be an absolutely convergent Fourier series. Then*

$$\frac{1}{M} \sum_{r=0}^{M-1} \Psi\left(u + \frac{r}{M}\right) = \sum_{\ell \in \mathbb{Z}} c_{\ell M} e^{2\pi i \ell M u}. \quad (63)$$

If $|c_k| \leq C e^{-\rho|k|}$, then

$$\left| \frac{1}{M} \sum_{r=0}^{M-1} \Psi\left(u + \frac{r}{M}\right) - c_0 \right| \leq \frac{2C e^{-\rho M}}{1 - e^{-\rho M}}. \quad (64)$$

Proof. Insert the Fourier series and use

$$\frac{1}{M} \sum_{r=0}^{M-1} e^{2\pi i k r / M} = \begin{cases} 1, & M \mid k, \\ 0, & M \nmid k. \end{cases}$$

The error bound is the geometric sum over nonzero multiples of M . □

The two filters answer different questions. Phase-preserving Lagrange interpolation recovers the limit while keeping all periodic coefficients fixed. Root-of-unity averaging projects a periodic coefficient onto the Fourier modes divisible by M , converging exponentially to its mean when the phase function is analytic in a strip.

9.3 Exact Lambert- W sample nodes

Suppose the relevant endpoint coordinate satisfies

$$x = (\lambda + \beta) e^{-a\lambda}, \quad (65)$$

with $a > 0$. Then

$$\boxed{\lambda = -\beta - \frac{1}{a} W_{-1}(-a x e^{-a\beta})}, \quad (66)$$

where the lower branch is selected for large positive λ . Conversely, once a starting phase $\lambda \equiv u \pmod{1}$ is chosen, the phase-locked physical nodes are

$$x_m = (\lambda + m + \beta) e^{-a(\lambda+m)}, \quad m = 0, \dots, p. \quad (67)$$

Their ratio is

$$\frac{x_{m+1}}{x_m} = q \frac{\lambda + m + 1 + \beta}{\lambda + m + \beta}, \quad (68)$$

so they are asymptotically geometric but carry exactly the correction needed to preserve the Lambert phase.

9.4 A practical composite algorithm

A numerically robust endpoint study can use the following sequence.

Algorithm 9.4 (Phase-resolved endpoint extraction). Given an observable $R(x)$ believed to have a periodic lower-Lambert expansion:

1. derive or fit the monotone saddle map $x \leftrightarrow \lambda$ and choose a phase $u \in [0, 1)$;
2. select a large $\lambda \equiv u \pmod{1}$ and generate x_m from (67);
3. evaluate $R(x_m)$ at increased precision;
4. apply the weights (58) in barycentric or compensated form;
5. repeat for several u to reconstruct the periodic coefficient functions;
6. optionally average M equally spaced phases and use (63) to isolate the mean or selected Fourier harmonics.

This is a synthesis of Lambert inversion, Lagrange interpolation, Richardson cancellation, and Fourier projection, but its exact weight formula and fixed-phase error theorem are not the same as the repository’s existing geometric-grid filter.

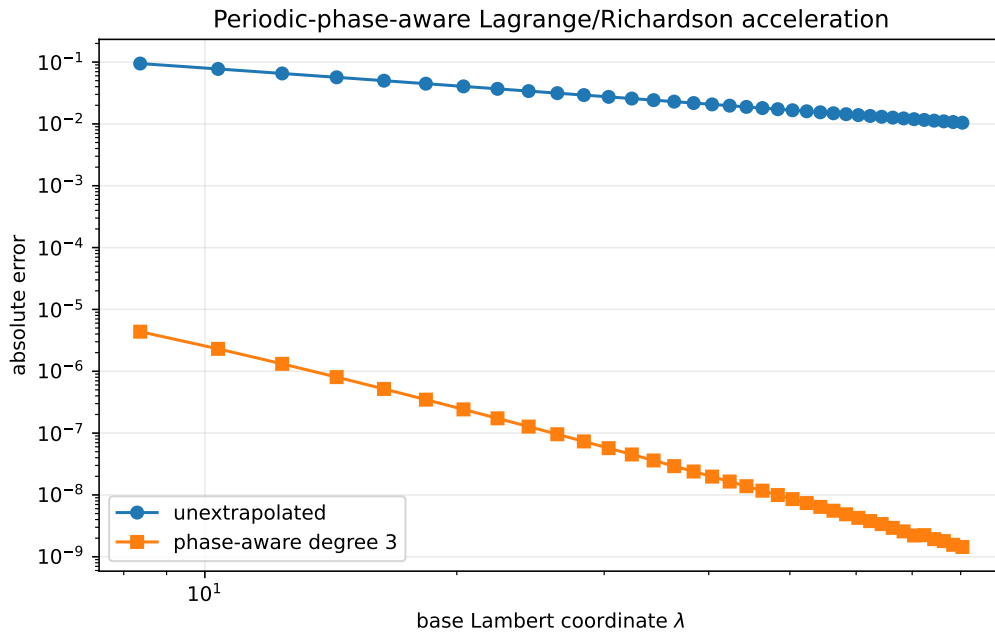


Figure 4: A synthetic periodic-phase model used only to verify [Theorem 9.1](#). The raw error has slope approximately -1 ; a degree-three phase-preserving filter has observed slope approximately -3.82 , approaching the predicted -4 . This experiment does not assume or fit the unknown endpoint coefficients of F_q .

10 Deterministic numerical experiments

The companion program `numerical_experiments.py` uses no Monte Carlo sampling. It is intended to test consequences of proved identities and to provide reconnaissance for the conjectures.

10.1 Fixed-point density solver

Starting from the uniform density, the code repeatedly applies the distributional fixed-point operator. On a uniform grid it computes a CDF by trapezoidal integration and then updates

$$f^{\text{new}}(x) = \frac{F^{\text{old}}(x/q) - F^{\text{old}}((x - c)/q)}{c}.$$

The density is renormalized after each step, and convergence is measured in the sup norm. The output verifies mass, symmetry, and the exact plateau. Representative results are shown in [Table 1](#).

Table 1: Fixed-point density diagnostics.

q	iterations	mass error	symmetry error	$f_q(1/2)$
0.30	11	$2.2 \cdot 10^{-16}$	$1.2 \cdot 10^{-15}$	1.42857142857
0.50	19	0	$4.2 \cdot 10^{-15}$	2.00000000000
0.75	45	0	$9.2 \cdot 10^{-14}$	3.47864814200
0.95	247	0	$2.6 \cdot 10^{-12}$	8.56266675023

For $q = 0.30$, the theoretical plateau height is $1/(1 - q) = 10/7$, and the maximum interior grid error on $[q, 1 - q]$ was exactly zero at machine precision.

10.2 Cumulant checks

The script computes standardized cumulants both from the general Bernoulli formula and from the simplified rational expressions. The differences were at most machine roundoff. For example,

q	σ_q^2	$\gamma_4(q)$	$\gamma_6(q)$
0.25	0.0500000	-1.0588235	5.6514914
0.50	0.0277778	-0.7200000	2.9387755
0.75	0.0119048	-0.3360000	0.6985447
0.90	0.0043860	-0.1259669	0.1003783
0.97	0.0012690	-0.0365397	0.0084745

The rapid decrease of higher standardized cumulants supports the Edgeworth program.

10.3 Endpoint bounds

For $q = 1/2$ and x from 10^{-4} down to 10^{-60} , the program evaluates the exact logarithms of the finite-simplex upper bound and balanced-product lower bound. After adding $L^2/(2a) + (L \log L)/a$ and dividing by L , both residuals remain bounded, as displayed in [Figure 3](#). This is a numerical visualization of a theorem rather than an independent fit.

10.4 Phase-aware extrapolation

The synthetic model is

$$R(\lambda) = L + \frac{a_1(u)}{\lambda} + \frac{a_2(u)}{\lambda^2} + \frac{a_3(u)}{\lambda^3} + \frac{a_4(u)}{\lambda^4}, \quad u = \{\lambda\},$$

with trigonometric coefficient functions. Sampling at $\lambda, \lambda + 1, \lambda + 2, \lambda + 3$ holds u fixed. A log-log regression gave raw slope -0.990967 and accelerated slope -3.824509 ; the latter tends to -4 as the asymptotic range and precision increase.

10.5 Zero multiplicities

For $b = 2, 3, 4, 5$ and $1 \leq |N| \leq 64$, the code enumerates vanishing sinc factors and verifies $\text{mult} = 1 + v_b(N)$ without exception. This check is elementary but guards against an easy normalization error in the lattice spacing.

10.6 Reproducibility

Running

```
python3 numerical_experiments.py
```

from the report directory regenerates the four PDF figures, a CSV summary, and the text log. The script contains detailed comments, parameter checks, deterministic stopping criteria, and no external data dependency.

11 Conjectures and promising research directions

This section collects the most promising next steps in a form designed to be falsifiable and, where possible, formalizable.

11.1 Analytic shape

Conjecture 11.1 (Strict log-concavity). *For $q > 1/2$, the density f_q is strictly log-concave on $(0, 1)$. Equivalently,*

$$f_q(\theta x + (1 - \theta)y) > f_q(x)^\theta f_q(y)^{1-\theta}$$

for distinct $x, y \in (0, 1)$ and $0 < \theta < 1$.

The exact plateau theorem shows that $q = 1/2$ is the only possible transition value. A proof would likely combine equality cases in Prékopa–Leindler with the self-similar decomposition and the fact that no single interval dominates the remaining support when $q > 1/2$.

Problem 11.2 (Monotonicity of the peak). *Prove or disprove that $q \mapsto f_q(1/2)$ is strictly increasing. Convex-order contraction suggests the statement but does not imply pointwise density order. The Gaussian prediction is*

$$f_q(1/2) \sim \sqrt{\frac{6(1+q)}{\pi(1-q)}}.$$

11.2 Uniform Gaussian asymptotics

Conjecture 11.3 (All-orders Edgeworth expansion). *The expansion in [Conjecture 6.2](#) holds uniformly with explicit coefficients generated by (35). In particular,*

$$f_q(1/2) = \frac{1}{\sqrt{2\pi}\sigma_q} \left(1 - \frac{3}{20}(1-q) + \mathcal{O}((1-q)^2) \right).$$

A promising proof strategy is to split the normalized Fourier integral at $|t| = (1-q)^{-\eta}$, use the cumulant expansion on the central range, and exploit a block of roughly $(1-q)^{-1}$ sinc factors for a q -uniform tail majorant.

11.3 Endpoint phase and inverse transfer

Conjecture 11.4 (Nonconstant periodic amplitude). *For generic q , the leading phase function $A_{q,0}$ in (50) is nonconstant, real analytic, and has no vanishing nonzero Fourier coefficient. Exceptional q , if any, form a thin arithmetic set.*

Conjecture 11.5 (Inverse phase transfer). *Let $Q_q = F_q^{-1}$. The periodic endpoint functions of F_q transfer to a complete expansion of $Q_q(y)$ through the lower branch W_{-1} and formal Lagrange inversion. The phase of the inverse is not simply the phase of $\log(1/y)$; it includes the derivative of the saddle map and a computable phase shift from the leading amplitude.*

The phase-aware sampling theorem provides a practical way to distinguish these possibilities numerically without averaging the phase away.

11.4 A two-parameter large-deviation/Gaussian phase diagram

The limits $x \downarrow 0$ and $q \uparrow 1$ do not commute. For fixed q , the endpoint law is quadratic in $\log(1/x)$. For $q \uparrow 1$ at the central scale $x = 1/2 + \sigma_q z$, the law is Gaussian. Between them lies a double-scaling region.

Problem 11.6 (Joint scaling). *Determine a uniform asymptotic for*

$$\mathbb{P}\left(X_q \leq \frac{1}{2} - r_q\right)$$

when $q \uparrow 1$ and $r_q/\sigma_q \rightarrow \infty$ while $r_q < 1/2$. Identify the transition from the Gaussian rate $r_q^2/(2\sigma_q^2)$ to the logarithmic endpoint action of Theorem 7.1. A natural tool is the exact cumulant generating product (19) and a saddle that moves through the geometric scale lattice.

11.5 Arithmetic zero collisions

Problem 11.7 (Algebraic collision graph). *For algebraic q , define a graph on scales $j \geq 0$ by joining j and k when $q^{|j-k|} \in \mathbb{Q}$. Express Fourier-zero multiplicities in terms of rational valuations along connected components. Determine when the multiplicity function is automatic or regular in the sense of Allouche–Shallit.*

11.6 Beyond binary Thue–Morse

Problem 11.8 (Contiguous completion in higher base). *For $b \geq 3$, find the minimal collection of digit factors $\prod_j P(z^{b^j})$ whose finite expansion is contiguous and whose coefficient sequence retains a signed Prouhet cancellation law. The obvious choice $P(z) = 1 + z + \dots + z^{b-1}$ loses signs; signed cyclotomic or character-valued factors may yield new automatic analogues.*

11.7 Numerical analysis and conditioning

Problem 11.9 (Stable phase-aware barycentric formula). *Derive a backward-stable barycentric implementation of Theorem 9.1 whose roundoff amplification is explicit in p and λ . Couple it to arbitrary-precision Lambert- W node generation, and optimize phase averaging against truncation and rounding errors.*

12 A Lean formalization roadmap

The new family can be developed in layers that mirror the existing repository architecture. The following order minimizes dependence on asymptotic machinery.

12.1 Probability layer

1. Define the geometric coefficient sequence $w_j(q) = (1 - q)q^j$ and prove summability, nonnegativity, and total mass one.
2. Name X_q and its law, then specialize the generic probability, carried-interval, reflection, continuous-linear naturality, and scalar affine-recursion theorems.
3. Prove equality of the topological support with $[0, 1]$, then derive the q -specific CDF integral equation, atomlessness or absolute continuity as needed, and the density refinement.
4. Derive characteristic-function products from finite products plus dominated convergence.

Since the pinned snapshot, items 1 and 2 have been formalized, in stronger pointwise form, by `WeightedUniformSeries.lean`, `WeightedUniformDistribution.lean`, and `GeometricUniformLaw.lean`. The law-reflection and independent-copy affine-refinement parts of item 3 are also formalized. Exact topological support and the CDF integral equation in item 3 remain open, as do atomlessness, the density consequences, fixed-point uniqueness, and the characteristic-function program in item 4.

12.2 Analysis layer

1. Formalize the fixed- N Fourier bound (11) and deduce smoothness by a reusable Fourier inversion theorem.
2. Prove log-concavity first for finite convolutions and then under the density limit.
3. Formalize the plateau interval by the extended-CDF refinement equation.
4. Separate strict unimodality from the stronger strict-log-concavity conjecture.

12.3 Algebraic layer

1. Prove the Bernoulli cumulant formula from the uniform mgf and a geometric series.
2. Connect moments to complete Bell polynomials.
3. Define $P_{q,n}$ by formal power series and prove the Appell, expectation-inverse, and refinement identities.
4. Import or prove the finite q -binomial theorem needed for (26).

12.4 Order and limit layer

1. Introduce the coefficient-vector majorization theorem in a form usable for infinite ℓ^1 sequences, then prove convex-order monotonicity.
2. Prove the triangular-array CLT by a bounded Lindeberg theorem.
3. Formalize standardized cumulants and the compact-frequency expansion.

12.5 Endpoint and numerical layer

1. Formalize the simplex-volume lemma with nonidentical coordinate scales.
2. Formalize the balanced product event and Stirling bounds sufficient for (43).
3. Prove the Lambert- W inversion corollary using monotonicity and the lower/upper branch API already present in the project.

4. Prove the phase-aware Lagrange weights as an exact rational-function identity before adding analytic remainder estimates.

The endpoint theorem is especially suitable for formalization because it avoids contour integrals: the only asymptotic ingredients are explicit integer choices, simplex volume, Stirling bounds, and elementary logarithmic estimates.

13 Conclusion

The dyadic Fabius–Rvachev theory is rigid enough that many formulas appear isolated until the geometric ratio is allowed to vary. The family

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j$$

places several of the repository’s central themes in a single parameter space. Bernoulli cumulants and Bell moments become exact for every q ; Gaussian-binomial coefficients encode the elementary symmetric functions of the geometric scales; Bernoulli polynomials invert each uniform-averaging factor; convex majorization orders the entire family; the dyadic value marks a genuine plateau-to-strict-unimodality transition; and $q \uparrow 1$ produces a Gaussian law with explicit cumulant corrections. At the opposite endpoint, a simple finite-dimensional argument already forces both the quadratic logarithmic action and the $L \log L$ entropy correction, while Lambert W inverts the resulting quantile scale.

The Fourier side exhibits an equally clean dichotomy. Reciprocal-integer ratios create valuation-controlled zero multiplicities, whereas generic irrational powers give simple zeros. Base two is singled out not merely by convention but because it is the unique base in which the signed subset product fills a contiguous integer block, thereby producing the Thue–Morse sequence. Finally, the periodic lower-Lambert corrections that complicate sharp endpoint numerics can be handled by sampling at integer-separated Lambert variables and applying an explicit Lagrange filter; phase averaging supplies the complementary Fourier projection.

The next frontier is to join these paper-level results into a uniform saddle theory in both q and x , determine the periodic coefficient functions, prove a global Edgeworth expansion, and classify algebraic zero collisions. Each problem is concrete enough to support numerical experiments and sufficiently structured to admit Lean formalization.

A Complete TeX corpus inventory

The following ledger was used to enforce corpus-relative nonduplication. It is also included as the standalone file `corpus_inventory.txt`.

```
Corpus inventory used for "Beyond the Dyadic Fabius Web"
Repository snapshot: VladimirReshetnikov/ProveIt, main, commit 24da985eba70e46d862acdadd476793a428213e
Inspection date: 2026-08-27
Scope: every *.tex file under Analysis/FabiusFunction/docs, recursively
Total: 57 files

A. Living and current synthesis documents (6)
1. Analysis/FabiusFunction/docs/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. Analysis/FabiusFunction/docs/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. Analysis/FabiusFunction/docs/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex

B. Current Fourier-decay draft and audit campaign (12)
7. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
```

```

11. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
    Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-
    infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/
    rvachev_up_fourier_decay.tex
14. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/
    rvachev_up_fourier_decay.tex
15. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/
    Rvachev_Up_Fourier_Decay_Final.tex
16. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/
    Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/
    rvachev_up_fourier_decay_final.tex
18. Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/
    rvachev_up_fourier_decay_final.tex

C. Included paper sources (7)
19. Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. Analysis/FabiusFunction/docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
22. Analysis/FabiusFunction/docs/papers/arXiv-1702.05442v1/09-Function.tex
23. Analysis/FabiusFunction/docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. Analysis/FabiusFunction/docs/papers/arXiv-2211.13570v1/document.tex
25. Analysis/FabiusFunction/docs/papers/ubiq15/ubiq15-corrected.tex

D. Frozen historical archive (32)
Early notes
26. Analysis/FabiusFunction/docs/archive/early-notes/Fabius Asymptotic/Fabius Asymptotic.tex
27. Analysis/FabiusFunction/docs/archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over the
    signed Thue-Morse sequence.tex

Standalone studies
28. Analysis/FabiusFunction/docs/archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/
    Non_Elementarity_of_the_Fabius_Function.tex
29. Analysis/FabiusFunction/docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. Analysis/FabiusFunction/docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

Primary exposition versions
31. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex

Primary supplements
37. Analysis/FabiusFunction/docs/archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
38. Analysis/FabiusFunction/docs/archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex

q-calculus
39. Analysis/FabiusFunction/docs/archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40. Analysis/FabiusFunction/docs/archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex

Lean walkthrough
41. Analysis/FabiusFunction/docs/archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42. Analysis/FabiusFunction/docs/archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex

Glossary
43. Analysis/FabiusFunction/docs/archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44. Analysis/FabiusFunction/docs/archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex

Research-frontier dossiers and atlas versions
45. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/
    Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/
    Fabius_Dyadic_Formulae_to_Asymptotics.tex
48. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
50. Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.
    tex
51. Analysis/FabiusFunction/docs/archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52. Analysis/FabiusFunction/docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.
    tex
53. Analysis/FabiusFunction/docs/archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.
    tex
54. Analysis/FabiusFunction/docs/archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
55. Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
56. Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-2/
    Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57. Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

Notes
- The living primary exposition identifies commit 24da985eba70e46d862acdadd476793a428213e as its 27 August 2026 snapshot.
- The archive README explains that byte-identical historical versions were pruned, so the archive is a provenance map rather than a
    naive copy of every prior path.
- The two AIF sources are OCR/original and corrected versions of the same paper; both were included because the request was source-
    file exhaustive.

```

B Auxiliary proof details

B.1 Schur-convexity of weighted i.i.d. sums

For completeness, let $U_1, \dots, U_N$ be i.i.d. and let Φ be convex. The map

$$H(a_1, \dots, a_N) = \mathbb{E} \Phi \left(\sum_{j=1}^N a_j U_j \right)$$

is convex because it is an expectation of a convex function of the coefficient vector, and it is symmetric because the U_j are exchangeable. Every symmetric convex function is Schur-convex. Equivalently, it suffices to check a single Robin-Hood transfer: replace (a_i, a_j) by

$$(\theta a_i + (1 - \theta) a_j, (1 - \theta) a_i + \theta a_j), \quad 0 \leq \theta \leq 1.$$

The new weighted pair is the conditional expectation of a random permutation of the old pair, and Jensen's inequality decreases the expectation of Φ . Iterating transfers and passing to ℓ^1 limits proves [Theorem 5.1](#).

B.2 A direct expansion behind the inverse law

Set $S = \sqrt{2aY}$ and $L_0 = S - \log S$. Then

$$L_0^2 + 2L_0 \log L_0 = S^2 + \mathcal{O}((\log S)^2).$$

The uncertainty in (46) is $\mathcal{O}(L_y) = \mathcal{O}(S)$, and the derivative of $L^2 + 2L \log L$ is $2L + 2 \log L + 2 \asymp S$. Therefore the uncertainty in L_y is bounded, which gives the explicit second form (48) without invoking the full asymptotic series of W .

B.3 Endpoint upper and lower constants

The proof of [Theorem 7.1](#) can be converted into explicit inequalities. There exist $x_0(q) > 0$ and constants $C_-(q), C_+(q)$ such that for $0 < x < x_0(q)$,

$$-\frac{L^2}{2a} - \frac{L \log L}{a} - C_-(q)L \leq \log F_q(x) \leq -\frac{L^2}{2a} - \frac{L \log L}{a} + C_+(q)L.$$

One obtains admissible, though nonoptimal, constants by using

$$n \log n - n \leq \log(n!) \leq n \log n - n + \log n + 1$$

for sufficiently large n , together with $|n_{\pm} - L/a| \leq C(q)$. The companion numerical plot evaluates the sharper uncompressed finite- n expressions rather than these coarse constants.

B.4 Phase-aware weights as finite differences

The weights (58) also admit the form

$$w_m^{(p)}(\lambda) = \frac{(-1)^p}{p!} (-1)^m \binom{p}{m} (\lambda + m)^p.$$

Thus the estimator is a normalized p th forward difference applied to $(\lambda + m)^p R(\lambda + m)$. The Lagrange interpretation is preferable for error analysis, but the finite-difference form may be easier to formalize algebraically.

C Reproducibility manifest

The ZIP archive accompanying this report contains:

- `fabius_frontier_report.tex` — the present source;
- `fabius_frontier_report.pdf` — the compiled report;
- `numerical_experiments.py` — deterministic, commented experiments;
- `q_family_densities.pdf`;
- `standardized_cumulants.pdf`;
- `endpoint_bound_residuals.pdf`;
- `phase_aware_extrapolation.pdf`;
- `numerical_summary.csv`;
- `experiment_output.txt`;
- `corpus_inventory.txt`;
- `README.txt`.

All numerical outputs are reproducible from the Python file with NumPy and Matplotlib.

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